

Bike Sharing Demand Forecasting - Model Evaluation Report

This report evaluates machine learning forecasting models developed for the Bike Sharing Dataset. The objective is to predict hourly bicycle rental demand ('cnt') using weather, seasonal, and temporal features to support operational planning and logistics optimization.

1. Forecasting Objective

The objective is to forecast hourly bike-sharing utilization, cnt, so the bicycle rental company can plan bicycle availability, staffing, and logistics. The forecast should support operational planning throughout the year and provide an interpretable error measure for business users.

2. Dataset Overview

The workflow uses the UCI Bike Sharing Dataset, specifically the hourly rental extract. The data includes timestamp-derived calendar fields, season, holiday and working-day flags, weather situation, temperature, humidity, windspeed, and rental counts. A chronological train/test split is used so evaluation resembles forecasting future demand.

- Target: cnt, total hourly rental count.
- Time coverage: historical hourly observations from 2011 and 2012.
- Excluded leakage fields: casual and registered, because they are components of cnt.

3. Modeling Approach

The project compares four regression approaches: Linear Regression, Random Forest, Gradient Boosting, and XGBoost. Each model is trained through a pipeline that includes preprocessing, so training and prediction use the same feature transformations.

- Linear Regression provides a simple baseline.
- Random Forest captures nonlinear interactions with bagged decision trees.
- Gradient Boosting captures nonlinear effects through sequential boosting.
- XGBoost is included as a high-performing boosted-tree model suitable for tabular demand forecasting.

4. Feature Engineering

The feature engineering process creates production-friendly model inputs. Calendar and weather fields are retained, categorical variables are encoded in model pipelines, and cyclical time features are added for hour and month.

- Time features: hour, month, weekday, working day, holiday, year, plus hour/month sine and cosine features.
- Weather features: weather situation, temperature, feeling temperature, humidity, and windspeed.
- Encoding: categorical features are one-hot encoded inside the pipeline.
- Scaling: numeric variables are scaled for Linear Regression and passed through for tree-based models.

5. Model Performance Comparison

The main comparison metric is mean absolute deviation, implemented as mean absolute error. This is the most useful metric for the planning team because it directly represents the average hourly bike-count miss.

Model	Mean absolute deviation	MAPE	R-squared
XGBoost	50.91	0.491	0.886
Random Forest	53.47	0.354	0.869
Gradient Boosting	57.00	0.608	0.860
Linear Regression	98.69	2.114	0.632

The supporting plots show model performance by MAD, MAPE, and R-squared, as well as forecast behavior over the latest test-period window.

6. Final Model Selection

The final recommended model is XGBoost. It achieved the lowest mean absolute deviation at 50.91 hourly rentals and the highest R-squared among the evaluated models. This makes it the most suitable business-case model in the current workflow.

- It captures nonlinear interactions between time, season, weather, and working-day behavior.
- It is accurate enough to support hourly operational planning with a clear error benchmark.
- It is packaged as a reusable model artifact that can be loaded by the Streamlit dashboard and batch prediction code.

7. Production Readiness

The code is structured for maintainability in a daily prediction service. Reusable Python modules handle ingestion, validation, feature engineering, model training, evaluation, plotting, report generation, scenario prediction, and the Streamlit UI.

- Automated tests validate data, feature engineering, evaluation summaries, and scenario input generation.
- Model artifacts are saved in the models folder for reproducible loading.
- Generated predictions, comparison metrics, and plots are stored in data/processed and reports/figures.
- The Streamlit app lets users inspect model comparison and test scenario predictions interactively.

8. Forecast Refresh Recommendations

Forecasts should be refreshed daily after the latest complete demand data is available. A daily refresh is appropriate because demand depends on calendar patterns and weather assumptions that can change from one day to the next.

- Run daily refreshes for regular planning, staffing, and bicycle readiness decisions.
- Run intraday refreshes when weather forecasts change materially or live demand deviates from forecast.
- Use the model mainly for short- to medium-term operational planning where weather and calendar inputs are reliable.
- For annual planning, combine model output with business assumptions such as fleet changes, station changes, pricing, holidays, and local events.

9. Conclusion

XGBoost is the recommended forecasting model for Part 1. It provides the best performance in the model comparison, with a mean absolute deviation of 50.91 hourly rentals. The result is decision-ready for operational planning when combined with production monitoring, regular refreshes, and business context for events and long-range planning.